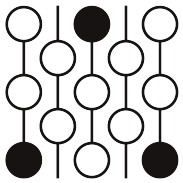
Review of Problem 5 – Astra

First Proof Third Batch
Reviewer 1
October 8–9, 2026



1 Review

The solution is mathematically correct. It provides a counterexample to the given question. The counterexample is constructed using an infinite cyclic projective plane, a construction studied in the literature. The new observation is to appropriately use this construction to give a counterexample to the question. This is very similar to the example constructed in submissions of other AI systems but different from the human submission. There are some important missing attributions. The exposition is not very good. An overview of the proof is not available, and many parts of the proof are unmotivated.

Correctness.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected
- 0: blank solution, or acknowledged partial result
- -0.5: major errors
- -1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
Score: 1

Exposition.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort
- -1: very poor exposition

Score: 0

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
- -1: blatant plagiarism or hallucinations

Score: -1

2 Submission with inline comments

Abstract

The proposed implication is false for general hyperfields. We construct a countably infinite counterexample whose multiplicative group is infinite cyclic. The construction uses an explicitly defined infinite difference set. All required incidence properties and all hyperfield axioms are verified directly.

3 Statement and conventions

We will prove the following.

Theorem 1. *There exists a hyperfield*

$$\mathbb{H} = \{0\} \sqcup \{t^n : n \in \mathbb{Z}\}, \qquad t^m t^n = t^{m+n},$$

in which every element is its own additive inverse and

$$0 \in 1 \boxplus t \boxplus t^6,$$

but

$$0 \notin 1 \boxplus t^3 \boxplus t^{18} \boxplus t^7 \boxplus t^7 \boxplus t^7.$$

Consequently, the implication in the question fails for

$$x = 1, \qquad y = t, \qquad z = t^6.$$

Here the symbols t^n are pairwise distinct formal symbols, and $1 = t^0$. In particular, the integer 0, when used as an exponent or an incidence point, must be distinguished from the hyperfield element 0.

For subsets A, B of a set equipped with hyperaddition, we use the convention

$$A \boxplus B = \bigcup_{\substack{a \in A \\ b \in B}} (a \boxplus b).$$

Singleton sets are identified with their elements in such expressions.

We use the usual definition of a commutative hyperfield: its nonzero elements form an abelian multiplicative group, zero is multiplicatively absorbing, and hyperaddition is a commutative, associative, nonempty-valued operation with neutral element 0, unique additive inverses, and reversibility

$$c \in a \boxplus b \quad \Longleftrightarrow \quad b \in c \boxplus (-a).$$

Multiplication must distribute over hyperaddition as an equality of sets.

The connection between projective incidence geometry and hyperfields extending the Krasner hyperfield is developed in [1, Sections 3–4]. The construction below, including the verification of the hyperfield axioms, is self-contained.

4 An infinite difference set

> **Reviewer 1**
>
> This section (and the introduction) gives no motivation for why infinite difference set is a relevant object to look at. The construction of this infinite difference set is essentially the argument in Theorem 3.1 of Marshall Hall's 1947 paper "Cyclic projective planes," which was never credited.

Lemma 2. *There exists an infinite set $D \subseteq \mathbb{Z}_{\geq 0}$ such that*

1. $$D \cap \{0, 1, \ldots, 38\} = \{0, 1, 6, 20, 23, 27\};$$

2. *every nonzero integer has a unique expression $d - e$, where $d, e \in D$.*

Proof. For a finite set $A \subseteq \mathbb{Z}_{\geq 0}$, put

$$\Delta(A) = \{a - b : a, b \in A,\ a > b\}.$$

Say that A has distinct positive differences if each element of $\Delta(A)$ arises from exactly one ordered pair (a, b).

Begin with

$$A_0 = \{0, 1, 6, 20, 23, 27\}.$$

Its positive differences are

$$1, 3, 4, 5, 6, 7, 14, 17, 19, 20, 21, 22, 23, 26, 27.$$

These are fifteen distinct integers, accounting for all $\binom{6}{2} = 15$ pairs. Thus A_0 has distinct positive differences.

We first describe an extension step. Suppose that A is a finite set containing A_0, that A has distinct positive differences, and that $n \geq 1$ does not belong to $\Delta(A)$. Set

$$M = \max A, \qquad u = 2M + n + 1, \qquad A' = A \cup \{u, u + n\}.$$

All positive differences in A' occur in the following four sets:

$$\Delta(A), \qquad \{n\}, \qquad \{u - a : a \in A\}, \qquad \{u + n - a : a \in A\}.$$

These sets are pairwise disjoint. Indeed, $n \notin \Delta(A)$, while every integer in either of the last two sets is at least

$$u - M = M + n + 1 > \max\{M, n\}.$$

Every element of $\Delta(A)$ is at most M. Finally, an equality

$$u - a = u + n - b$$

would imply $b - a = n$, contrary to $n \notin \Delta(A)$. Within each of the last two sets, distinct elements of A give distinct differences. Therefore A' again has distinct positive differences. Moreover, $n \in \Delta(A')$.

Now define A_n recursively for $n \geq 1$. If $n \in \Delta(A_{n-1})$, let $A_n = A_{n-1}$. Otherwise apply the extension step with $A = A_{n-1}$ and this value of n. Set

$$D = \bigcup_{n \geq 0} A_n.$$

Every positive integer n belongs to $\Delta(A_n)$, hence is a difference of two elements of D. Such a representation is unique: any two competing representations would have all their entries in some

finite A_N, contradicting the distinct-difference property of A_N. Reversing the ordered pair proves existence and uniqueness for negative integers as well.

At every extension step $M \geq 27$, so

$$u = 2M + n + 1 \geq 56 > 38.$$

Thus no new element at most 38 is ever introduced, proving the first assertion. Finally, D is infinite, since a finite set has only finitely many differences. □

Lemma 3. *For $k \in \mathbb{Z}$, define*

$$L_k = D + k = \{d + k : d \in D\}.$$

Regard the integers as points and the sets L_k as lines. Then:

1. *every two distinct points lie on exactly one line;*
2. *every two distinct lines meet in exactly one point;*
3. *every line is infinite;*
4. *integer translations preserve the collection of lines.*

> **Reviewer 1**
>
> The ideas and proof of this lemma appears in "Projective Geometries as Multigroups" by Prenowitz, "A Theorem in Finite Projective Geometry and Some Applications to Number Theory" by Singer, and should have been given some attributions.

Proof. Let $p \neq q$ be integers. By Lemma 2, there is a unique ordered pair $d, e \in D$ satisfying

$$p - q = d - e.$$

Then

$$k = p - d = q - e$$

is the unique integer for which $p, q \in L_k$. This proves the first assertion.

For $k \neq \ell$, a point of $L_k \cap L_\ell$ corresponds precisely to an ordered pair $d, e \in D$ satisfying

$$k + d = \ell + e, \qquad \text{equivalently} \qquad d - e = \ell - k.$$

There is exactly one such ordered pair, so the intersection consists of exactly one point. Since each L_k is infinite, this also shows that different indices give different lines.

The third assertion follows from the infinitude of D, and the fourth from

$$L_k + s = L_{k+s}.$$

□

For distinct integers m, n, write $L(m, n)$ for their unique common line.

5 Construction and verification of the hyperfield

> **Reviewer 1**
>
> This section should explicitly cite [1] and the 1943 paper "Projective Geometries as Multigroups" by Prenowitz. [1] was briefly mentioned (very cryptically) in the introduction, but it should also be mentioned here. The proof here is very similar to the proof in [1], and could have been omitted.

Let

$$\mathbb{H} = \{0\} \sqcup \{t^n : n \in \mathbb{Z}\},$$

with multiplication

$$t^m t^n = t^{m+n}, \qquad 0a = a0 = 0.$$

For distinct nonzero elements $a = t^m$ and $b = t^n$, define their line by

$$\Lambda(a, b) = \{t^r : r \in L(m, n)\} \subseteq \mathbb{H}^\times.$$

The incidence properties of Lemma 3 hold equally for these lines in $\mathbb{H}^\times$.

Define hyperaddition by

$$0 \boxplus a = a \boxplus 0 = \{a\}$$

and, for $a, b \neq 0$, by

$$a \boxplus b = \begin{cases} \{0, a\}, & a = b, \\ \Lambda(a, b) \setminus \{a, b\}, & a \neq b. \end{cases} \tag{1}$$

Proposition 4. *These operations make $\mathbb{H}$ a commutative hyperfield. Its additive negation is the identity map.*

Proof. The nonzero elements form an infinite cyclic multiplicative group with identity $1 = t^0$. Zero is absorbing by definition.

Nonempty sums, commutativity, and the neutral element. Every line is infinite, so all sums in (1) are nonempty. Commutativity and the additive neutral-element axiom follow directly from the definition.

Additive inverses and reversibility. For all $a, b \in \mathbb{H}$, the definition gives

$$0 \in a \boxplus b \quad \iff \quad a = b. \tag{2}$$

Consequently, every element has a unique additive inverse, namely

$$-a = a.$$

To verify reversibility, first suppose that a, b, c are all nonzero. Then $c \in a \boxplus b$ holds precisely when either all three elements coincide, or they are pairwise distinct and lie on a common line. Both conditions are symmetric in a, b, c. If at least one element is zero, membership holds precisely when the other two elements are equal, again a symmetric condition. Hence

$$c \in a \boxplus b \quad \iff \quad b \in c \boxplus a \quad \iff \quad b \in c \boxplus (-a),$$

as required.

Associativity. If one of the three elements is zero, associativity follows from the neutral-element rule. We therefore consider nonzero elements.

If all three elements equal a, then

$$(a \boxplus a) \boxplus a = \{0, a\} \boxplus a = \{0, a\}.$$

Suppose next that the multiset of the three elements is $\{a, a, b\}$, where $a \neq b$, and let $\Lambda = \Lambda(a, b)$. Then

$$(a \boxplus a) \boxplus b = \{b\} \cup \big(\Lambda \setminus \{a, b\}\big) = \Lambda \setminus \{a\}.$$

Also, uniqueness of the line through two distinct points gives

$$\begin{aligned}(a \boxplus b) \boxplus a &= \bigcup_{s \in \Lambda \setminus \{a,b\}} \big(\Lambda \setminus \{s, a\}\big) \\ &= \Lambda \setminus \{a\}.\end{aligned}$$

For the last equality, given $v \in \Lambda \setminus \{a\}$, choose $s \in \Lambda \setminus \{a, b, v\}$, which is possible because Λ is infinite.

Now suppose that a, b, c are pairwise distinct and lie on one line Λ. We claim that

$$(a \boxplus b) \boxplus c = \{0\} \cup \Lambda. \tag{3}$$

Every summand on the left is contained in $\{0\} \cup \Lambda$. Choosing the intermediate element $c \in a \boxplus b$ produces both 0 and c. Given $v \in \Lambda \setminus \{c\}$, choose

$$s \in \Lambda \setminus \{a, b, c, v\}.$$

Then $s \in a \boxplus b$ and $v \in s \boxplus c$, proving (3).

Finally, suppose that a, b, c are noncollinear. In this case

$$(a \boxplus b) \boxplus c = \mathbb{H}^\times \setminus \big(\Lambda(a, b) \cup \Lambda(a, c) \cup \Lambda(b, c)\big). \tag{4}$$

To prove the forward inclusion, take

$$s \in \Lambda(a, b) \setminus \{a, b\}.$$

Since $c \notin \Lambda(a, b)$, the elements s, c are distinct. The line $\Lambda(s, c)$ meets $\Lambda(a, b)$ only at s, and meets each of $\Lambda(a, c)$ and $\Lambda(b, c)$ only at c. Therefore

$$s \boxplus c = \Lambda(s, c) \setminus \{s, c\}$$

avoids all three lines on the right of (4), as well as the hyperfield zero.

Conversely, let v lie outside those three lines. The line $\Lambda(c, v)$ meets $\Lambda(a, b)$ in a unique point s. This point is neither a nor b, since otherwise v would belong to $\Lambda(a, c)$ or $\Lambda(b, c)$. Also $v \neq s, c$. Thus

$$s \in a \boxplus b, \qquad v \in s \boxplus c,$$

proving the reverse inclusion.

The formulas just established are independent of the order in which the three elements are listed. Together with commutativity, they imply

$$(a \boxplus b) \boxplus c = a \boxplus (b \boxplus c)$$

in every case. Associativity for subsets follows by taking unions over their elements; hence finite hypersums can be parenthesized freely.

Distributivity. Multiplication by t^k acts on nonzero elements by translating their exponents by k. By Lemma 3, it carries lines to lines, and in particular

$$t^k \Lambda(t^m, t^n) = \Lambda(t^{m+k}, t^{n+k}) \qquad (m \neq n).$$

It follows from (1), including its equal-argument and zero-argument cases, that

$$t^k(a \boxplus b) = t^k a \boxplus t^k b.$$

For multiplication by zero, both sides are $\{0\}$, since $a \boxplus b$ is nonempty. Thus distributivity holds for all scalars. All hyperfield axioms have now been verified. $\square$

6 Failure of the cubic implication

Proof of Theorem 1. Use the hyperfield constructed above and set

$$x = 1, \qquad y = t, \qquad z = t^6.$$

The integers $0, 1, 6$ all belong to D. Consequently,

$$t^6 \in \Lambda(1, t) \setminus \{1, t\} = 1 \boxplus t.$$

It follows that

$$0 \in t^6 \boxplus t^6 \subseteq (1 \boxplus t) \boxplus t^6.$$

Thus the hypothesis in the question is satisfied.

Consider the line

$$\Lambda = \{t^r : r \in D - 20\}.$$

Because $20, 23, 27 \in D$, this line contains

$$1, \qquad t^3, \qquad t^7.$$

On the other hand,

$$t^{18} \notin \Lambda,$$

because $18 \in D - 20$ would imply $38 \in D$, contrary to Lemma 2.

Write

$$u = 1, \qquad v = t^3, \qquad w = t^{18}, \qquad q = t^7, \qquad A = u \boxplus v \boxplus w.$$

The points u, v, w are noncollinear, and $\Lambda(u, v) = \Lambda$. Formula (4) therefore gives

$$A \subseteq \mathbb{H}^\times \setminus \Lambda.$$

In particular,

$$0 \notin A, \qquad q \notin A. \tag{5}$$

The repeated sum of q satisfies

$$q \boxplus q \boxplus q = \{0, q\} \boxplus q = \{0, q\}.$$

Hence

$$\begin{aligned} u \boxplus v \boxplus w \boxplus q \boxplus q \boxplus q &= A \boxplus \{0, q\} \\ &= A \cup \bigcup_{a \in A} (a \boxplus q). \end{aligned}$$

The first set on the last line does not contain zero by (5). By (2), a set $a \boxplus q$ contains zero exactly when $a = q$; this is also excluded by (5). Therefore

$$0 \notin 1 \boxplus t^3 \boxplus t^{18} \boxplus t^7 \boxplus t^7 \boxplus t^7.$$

Finally, multiplication and negation in this hyperfield give

$$x^3 = 1, \qquad y^3 = t^3, \qquad z^3 = t^{18}, \qquad -xyz = xyz = t^7.$$

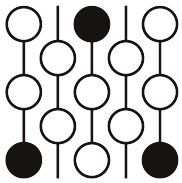
Thus the last displayed nonmembership is precisely the failure of the conclusion asked about in the question. □

References

[1] A. Connes and C. Consani, *The hyperring of adèle classes*, Journal of Number Theory **131** (2011), no. 2, 159–194.

Review of Problem 5 – Astra

First Proof Third Batch
Reviewer 2
October 8–9, 2026



1 Review

This problem turns out to admit a solution using existing work in the literature. Essentially, a solution may be found by combining a paper of Connes & Consani and a paper of Hall; the former of which is directly about hyperfields. Once these two papers have been found, no great effort is required to solve the problem, more on the level of a challenging homework problem (perhaps 2 pages once definitions have been given) than a research question. The review should be read in that light.

This submission successfully produces a correct counterexample to the cubic implication for hyperfields. I found no mathematical issues. The exposition is okay overall, I did not struggle to read it. There are some bad notational choices, but nothing egregious. The argument order harms the motivation in places, e.g. why do we construct an infinite difference set in section 4? This is not revealed to the reader until later. The citations are quite bad. The submission does include a reference to Connes & Consani for the idea, but only in one place, and not really explaining how the ideas in that paper are related to those here. A method of Hall for constructing difference sets from Sidon sets is used without any citation.

Correctness.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected
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- -1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
Score: 1

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Score: 0

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
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Score: 0

2 Submission with inline comments

Abstract

The proposed implication is false for general hyperfields. We construct a countably infinite counterexample whose multiplicative group is infinite cyclic. The construction uses an explicitly defined infinite difference set. All required incidence properties and all hyperfield axioms are verified directly.

3 Statement and conventions

We will prove the following.

Theorem 1. *There exists a hyperfield*

$$\mathbb{H} = \{0\} \sqcup \{t^n : n \in \mathbb{Z}\}, \qquad t^m t^n = t^{m+n},$$

in which every element is its own additive inverse and

$$0 \in 1 \boxplus t \boxplus t^6,$$

but

$$0 \notin 1 \boxplus t^3 \boxplus t^{18} \boxplus t^7 \boxplus t^7 \boxplus t^7.$$

Consequently, the implication in the question fails for

$$x = 1, \qquad y = t, \qquad z = t^6.$$

Here the symbols t^n are pairwise distinct formal symbols, and $1 = t^0$. In particular, the integer 0, when used as an exponent or an incidence point, must be distinguished from the hyperfield element 0.

For subsets A, B of a set equipped with hyperaddition, we use the convention

$$A \boxplus B = \bigcup_{\substack{a \in A \\ b \in B}} (a \boxplus b).$$

Singleton sets are identified with their elements in such expressions.

We use the usual definition of a commutative hyperfield: its nonzero elements form an abelian multiplicative group, zero is multiplicatively absorbing, and hyperaddition is a commutative, associative, nonempty-valued operation with neutral element 0, unique additive inverses, and reversibility

$$c \in a \boxplus b \quad \Longleftrightarrow \quad b \in c \boxplus (-a).$$

Multiplication must distribute over hyperaddition as an equality of sets.

> **Reviewer 2**
>
> I think it's a pretty poor choice to give the hyperfield axioms in paragraph form, makes it harder for the reader to check if the below example suffices.

The connection between projective incidence geometry and hyperfields extending the Krasner hyperfield is developed in [1, Sections 3–4]. The construction below, including the verification of the hyperfield axioms, is self-contained.

> **Reviewer 2**
>
> This is a pretty insufficient citation given that it's the key idea being exploited here.

4 An infinite difference set

> **Reviewer 2**
>
> This is an argument of Hall and should be cited.

Lemma 2. *There exists an infinite set $D \subseteq \mathbb{Z}_{\geq 0}$ such that*

1. $$D \cap \{0, 1, \ldots, 38\} = \{0, 1, 6, 20, 23, 27\};$$

2. *every nonzero integer has a unique expression $d - e$, where $d, e \in D$.*

Proof. For a finite set $A \subseteq \mathbb{Z}_{\geq 0}$, put

$$\Delta(A) = \{a - b : a, b \in A,\ a > b\}.$$

Say that A has distinct positive differences if each element of $\Delta(A)$ arises from exactly one ordered pair (a, b).

Begin with

$$A_0 = \{0, 1, 6, 20, 23, 27\}.$$

Its positive differences are

$$1, 3, 4, 5, 6, 7, 14, 17, 19, 20, 21, 22, 23, 26, 27.$$

These are fifteen distinct integers, accounting for all $\binom{6}{2} = 15$ pairs. Thus A_0 has distinct positive differences.

We first describe an extension step. Suppose that A is a finite set containing A_0, that A has distinct positive differences, and that $n \geq 1$ does not belong to $\Delta(A)$. Set

$$M = \max A, \qquad u = 2M + n + 1, \qquad A' = A \cup \{u, u + n\}.$$

All positive differences in A' occur in the following four sets:

$$\Delta(A), \qquad \{n\}, \qquad \{u - a : a \in A\}, \qquad \{u + n - a : a \in A\}.$$

These sets are pairwise disjoint. Indeed, $n \notin \Delta(A)$, while every integer in either of the last two sets is at least

$$u - M = M + n + 1 > \max\{M, n\}.$$

Every element of $\Delta(A)$ is at most M. Finally, an equality

$$u - a = u + n - b$$

would imply $b - a = n$, contrary to $n \notin \Delta(A)$. Within each of the last two sets, distinct elements of A give distinct differences. Therefore A' again has distinct positive differences. Moreover, $n \in \Delta(A')$.

Now define A_n recursively for $n \geq 1$. If $n \in \Delta(A_{n-1})$, let $A_n = A_{n-1}$. Otherwise apply the extension step with $A = A_{n-1}$ and this value of n. Set

$$D = \bigcup_{n \geq 0} A_n.$$

Every positive integer n belongs to $\Delta(A_n)$, hence is a difference of two elements of D. Such a representation is unique: any two competing representations would have all their entries in some finite A_N, contradicting the distinct-difference property of A_N. Reversing the ordered pair proves existence and uniqueness for negative integers as well.

At every extension step $M \geq 27$, so

$$u = 2M + n + 1 \geq 56 > 38.$$

Thus no new element at most 38 is ever introduced, proving the first assertion. Finally, D is infinite, since a finite set has only finitely many differences. □

> **Reviewer 2**
>
> This is a semi-standard way to define the axioms of a projective geometry and should be prefaced as such.

Lemma 3. *For $k \in \mathbb{Z}$, define*

$$L_k = D + k = \{d + k : d \in D\}.$$

Regard the integers as points and the sets L_k as lines. Then:

1. *every two distinct points lie on exactly one line;*
2. *every two distinct lines meet in exactly one point;*
3. *every line is infinite;*
4. *integer translations preserve the collection of lines.*

Proof. Let $p \neq q$ be integers. By Lemma 2, there is a unique ordered pair $d, e \in D$ satisfying

$$p - q = d - e.$$

Then

$$k = p - d = q - e$$

is the unique integer for which $p, q \in L_k$. This proves the first assertion.

For $k \neq \ell$, a point of $L_k \cap L_\ell$ corresponds precisely to an ordered pair $d, e \in D$ satisfying

$$k + d = \ell + e, \qquad \text{equivalently} \qquad d - e = \ell - k.$$

There is exactly one such ordered pair, so the intersection consists of exactly one point. Since each L_k is infinite, this also shows that different indices give different lines.

The third assertion follows from the infinitude of D, and the fourth from

$$L_k + s = L_{k+s}.$$

□

For distinct integers m, n, write $L(m, n)$ for their unique common line.

5 Construction and verification of the hyperfield

Let

$$\mathbb{H} = \{0\} \sqcup \{t^n : n \in \mathbb{Z}\},$$

with multiplication

$$t^m t^n = t^{m+n}, \qquad 0a = a0 = 0.$$

For distinct nonzero elements $a = t^m$ and $b = t^n$, define their line by

$$\Lambda(a, b) = \{t^r : r \in L(m, n)\} \subseteq \mathbb{H}^\times.$$

The incidence properties of Lemma 3 hold equally for these lines in $\mathbb{H}^\times$.

Define hyperaddition by

$$0 \boxplus a = a \boxplus 0 = \{a\}$$

and, for $a, b \neq 0$, by

$$a \boxplus b = \begin{cases} \{0, a\}, & a = b, \\ \Lambda(a, b) \setminus \{a, b\}, & a \neq b. \end{cases} \tag{1}$$

Proposition 4. *These operations make $\mathbb{H}$ a commutative hyperfield. Its additive negation is the identity map.*

Proof. The nonzero elements form an infinite cyclic multiplicative group with identity $1 = t^0$. Zero is absorbing by definition.

Nonempty sums, commutativity, and the neutral element. Every line is infinite, so all sums in (1) are nonempty. Commutativity and the additive neutral-element axiom follow directly from the definition.

Additive inverses and reversibility. For all $a, b \in \mathbb{H}$, the definition gives

$$0 \in a \boxplus b \quad \iff \quad a = b. \tag{2}$$

Consequently, every element has a unique additive inverse, namely

$$-a = a.$$

To verify reversibility, first suppose that a, b, c are all nonzero. Then $c \in a \boxplus b$ holds precisely when either all three elements coincide, or they are pairwise distinct and lie on a common line. Both conditions are symmetric in a, b, c. If at least one element is zero, membership holds precisely when the other two elements are equal, again a symmetric condition. Hence

$$c \in a \boxplus b \quad \iff \quad b \in c \boxplus a \quad \iff \quad b \in c \boxplus (-a),$$

as required.

Associativity. If one of the three elements is zero, associativity follows from the neutral-element rule. We therefore consider nonzero elements.

If all three elements equal a, then

$$(a \boxplus a) \boxplus a = \{0, a\} \boxplus a = \{0, a\}.$$

Suppose next that the multiset of the three elements is $\{a, a, b\}$, where $a \neq b$, and let $\Lambda = \Lambda(a, b)$. Then

$$(a \boxplus a) \boxplus b = \{b\} \cup \big(\Lambda \setminus \{a, b\}\big) = \Lambda \setminus \{a\}.$$

Also, uniqueness of the line through two distinct points gives

$$\begin{aligned}(a \boxplus b) \boxplus a &= \bigcup_{s \in \Lambda \setminus \{a,b\}} \big(\Lambda \setminus \{s, a\}\big) \\ &= \Lambda \setminus \{a\}.\end{aligned}$$

For the last equality, given $v \in \Lambda \setminus \{a\}$, choose $s \in \Lambda \setminus \{a, b, v\}$, which is possible because Λ is infinite.

Now suppose that a, b, c are pairwise distinct and lie on one line Λ. We claim that

$$(a \boxplus b) \boxplus c = \{0\} \cup \Lambda. \tag{3}$$

Every summand on the left is contained in $\{0\} \cup \Lambda$. Choosing the intermediate element $c \in a \boxplus b$ produces both 0 and c. Given $v \in \Lambda \setminus \{c\}$, choose

$$s \in \Lambda \setminus \{a, b, c, v\}.$$

Then $s \in a \boxplus b$ and $v \in s \boxplus c$, proving (3).

Finally, suppose that a, b, c are noncollinear. In this case

$$(a \boxplus b) \boxplus c = \mathbb{H}^\times \setminus \big(\Lambda(a, b) \cup \Lambda(a, c) \cup \Lambda(b, c)\big). \tag{4}$$

To prove the forward inclusion, take

$$s \in \Lambda(a, b) \setminus \{a, b\}.$$

Since $c \notin \Lambda(a, b)$, the elements s, c are distinct. The line $\Lambda(s, c)$ meets $\Lambda(a, b)$ only at s, and meets each of $\Lambda(a, c)$ and $\Lambda(b, c)$ only at c. Therefore

$$s \boxplus c = \Lambda(s, c) \setminus \{s, c\}$$

avoids all three lines on the right of (4), as well as the hyperfield zero.

Conversely, let v lie outside those three lines. The line $\Lambda(c, v)$ meets $\Lambda(a, b)$ in a unique point s. This point is neither a nor b, since otherwise v would belong to $\Lambda(a, c)$ or $\Lambda(b, c)$. Also $v \neq s, c$. Thus

$$s \in a \boxplus b, \qquad v \in s \boxplus c,$$

proving the reverse inclusion.

The formulas just established are independent of the order in which the three elements are listed. Together with commutativity, they imply

$$(a \boxplus b) \boxplus c = a \boxplus (b \boxplus c)$$

in every case. Associativity for subsets follows by taking unions over their elements; hence finite hypersums can be parenthesized freely.

Distributivity. Multiplication by t^k acts on nonzero elements by translating their exponents by k. By Lemma 3, it carries lines to lines, and in particular

$$t^k \Lambda(t^m, t^n) = \Lambda(t^{m+k}, t^{n+k}) \qquad (m \neq n).$$

It follows from (1), including its equal-argument and zero-argument cases, that

$$t^k(a \boxplus b) = t^k a \boxplus t^k b.$$

For multiplication by zero, both sides are $\{0\}$, since $a \boxplus b$ is nonempty. Thus distributivity holds for all scalars. All hyperfield axioms have now been verified. $\square$

6 Failure of the cubic implication

Proof of Theorem 1. Use the hyperfield constructed above and set

$$x = 1, \qquad y = t, \qquad z = t^6.$$

The integers $0, 1, 6$ all belong to D. Consequently,

$$t^6 \in \Lambda(1, t) \setminus \{1, t\} = 1 \boxplus t.$$

It follows that

$$0 \in t^6 \boxplus t^6 \subseteq (1 \boxplus t) \boxplus t^6.$$

Thus the hypothesis in the question is satisfied.

Consider the line

$$\Lambda = \{t^r : r \in D - 20\}.$$

Because $20, 23, 27 \in D$, this line contains

$$1, \qquad t^3, \qquad t^7.$$

On the other hand,

$$t^{18} \notin \Lambda,$$

because $18 \in D - 20$ would imply $38 \in D$, contrary to Lemma 2.

Write

$$u = 1, \qquad v = t^3, \qquad w = t^{18}, \qquad q = t^7, \qquad A = u \boxplus v \boxplus w.$$

The points u, v, w are noncollinear, and $\Lambda(u, v) = \Lambda$. Formula (4) therefore gives

$$A \subseteq \mathbb{H}^\times \setminus \Lambda.$$

In particular,

$$0 \notin A, \qquad q \notin A. \tag{5}$$

The repeated sum of q satisfies

$$q \boxplus q \boxplus q = \{0, q\} \boxplus q = \{0, q\}.$$

Hence

$$\begin{aligned} u \boxplus v \boxplus w \boxplus q \boxplus q \boxplus q &= A \boxplus \{0, q\} \\ &= A \cup \bigcup_{a \in A} (a \boxplus q). \end{aligned}$$

The first set on the last line does not contain zero by (5). By (2), a set $a \boxplus q$ contains zero exactly when $a = q$; this is also excluded by (5). Therefore

$$0 \notin 1 \boxplus t^3 \boxplus t^{18} \boxplus t^7 \boxplus t^7 \boxplus t^7.$$

Finally, multiplication and negation in this hyperfield give

$$x^3 = 1, \qquad y^3 = t^3, \qquad z^3 = t^{18}, \qquad -xyz = xyz = t^7.$$

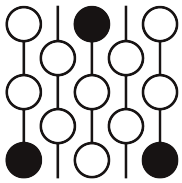
Thus the last displayed nonmembership is precisely the failure of the conclusion asked about in the question. □

References

[1] A. Connes and C. Consani, *The hyperring of adèle classes*, Journal of Number Theory **131** (2011), no. 2, 159–194.

Review of Problem 5 – Astra

First Proof Third Batch
Reviewer 3
October 8–9, 2026



1 Review

In a ring or a field, the identity $x + y + z = 0$ implies the identity $x^3 + y^3 + z^3 - 3xyz = 0$. This submitted note provides a counterexample for its counterpart for hyperfields, in which the addition is possibly multi-valued.

The counterexample presented in this note follows the ideas of Hall, Prenowitz, and Connes–Consani. Concretely, Prenowitz and Connes–Consani connected hyperfields with point-line arrangements, and Hall gave constructions of an infinite point-line arrangement from $\mathbb{Z}$. The proofs written down are all correct. However, the historical remarks are only at the end of Section 3 but not in the main body of the submission, and they are very brief, acknowledging only the work of Connes–Consani. The extension construction of the point-line arrangements (Section 4) is almost identical to the construction and proof in Hall, and the construction of a hyperfield from a projective plane (Section 5) is almost identical to the construction and proof in Connes–Consani.

For publication in a journal, I recommend a major revision. In particular, the author should remove the proofs that are already written in the literature and only cite the corresponding papers instead.

Correctness.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected
- 0: blank solution, or acknowledged partial result
- -0.5: major errors
- -1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
Score: 1

Exposition.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort
- -1: very poor exposition

Score: -1

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
- -1: blatant plagiarism or hallucinations

Score: -1

2 Submission with inline comments

Abstract

The proposed implication is false for general hyperfields. We construct a countably infinite counterexample whose multiplicative group is infinite cyclic. The construction uses an explicitly defined infinite difference set. All required incidence properties and all hyperfield axioms are verified directly.

3 Statement and conventions

We will prove the following.

Theorem 1. *There exists a hyperfield*

$$\mathbb{H} = \{0\} \sqcup \{t^n : n \in \mathbb{Z}\}, \qquad t^m t^n = t^{m+n},$$

in which every element is its own additive inverse and

$$0 \in 1 \boxplus t \boxplus t^6,$$

but

$$0 \notin 1 \boxplus t^3 \boxplus t^{18} \boxplus t^7 \boxplus t^7 \boxplus t^7.$$

Consequently, the implication in the question fails for

$$x = 1, \qquad y = t, \qquad z = t^6.$$

Here the symbols t^n are pairwise distinct formal symbols, and $1 = t^0$. In particular, the integer 0, when used as an exponent or an incidence point, must be distinguished from the hyperfield element 0.

For subsets A, B of a set equipped with hyperaddition, we use the convention

$$A \boxplus B = \bigcup_{\substack{a \in A \\ b \in B}} (a \boxplus b).$$

Singleton sets are identified with their elements in such expressions.

We use the usual definition of a commutative hyperfield: its nonzero elements form an abelian multiplicative group, zero is multiplicatively absorbing, and hyperaddition is a commutative, associative, nonempty-valued operation with neutral element 0, unique additive inverses, and reversibility

$$c \in a \boxplus b \quad \Longleftrightarrow \quad b \in c \boxplus (-a).$$

Multiplication must distribute over hyperaddition as an equality of sets.

> **Reviewer 3**
>
> The definition of hyperfields should be made clearer.

The connection between projective incidence geometry and hyperfields extending the Krasner hyperfield is developed in [1, Sections 3–4]. The construction below, including the verification of the hyperfield axioms, is self-contained.

4 An infinite difference set

Lemma 2. *There exists an infinite set $D \subseteq \mathbb{Z}_{\geq 0}$ such that*

1. $$D \cap \{0, 1, \ldots, 38\} = \{0, 1, 6, 20, 23, 27\};$$

2. *every nonzero integer has a unique expression $d - e$, where $d, e \in D$.*

Proof. For a finite set $A \subseteq \mathbb{Z}_{\geq 0}$, put

$$\Delta(A) = \{a - b : a, b \in A,\ a > b\}.$$

Say that A has distinct positive differences if each element of $\Delta(A)$ arises from exactly one ordered pair (a, b).

Begin with

$$A_0 = \{0, 1, 6, 20, 23, 27\}.$$

Its positive differences are

$$1, 3, 4, 5, 6, 7, 14, 17, 19, 20, 21, 22, 23, 26, 27.$$

These are fifteen distinct integers, accounting for all $\binom{6}{2} = 15$ pairs. Thus A_0 has distinct positive differences.

We first describe an extension step. Suppose that A is a finite set containing A_0, that A has distinct positive differences, and that $n \geq 1$ does not belong to $\Delta(A)$. Set

$$M = \max A, \qquad u = 2M + n + 1, \qquad A' = A \cup \{u, u + n\}.$$

All positive differences in A' occur in the following four sets:

$$\Delta(A), \qquad \{n\}, \qquad \{u - a : a \in A\}, \qquad \{u + n - a : a \in A\}.$$

These sets are pairwise disjoint. Indeed, $n \notin \Delta(A)$, while every integer in either of the last two sets is at least

$$u - M = M + n + 1 > \max\{M, n\}.$$

Every element of $\Delta(A)$ is at most M. Finally, an equality

$$u - a = u + n - b$$

would imply $b - a = n$, contrary to $n \notin \Delta(A)$. Within each of the last two sets, distinct elements of A give distinct differences. Therefore A' again has distinct positive differences. Moreover, $n \in \Delta(A')$.

Now define A_n recursively for $n \geq 1$. If $n \in \Delta(A_{n-1})$, let $A_n = A_{n-1}$. Otherwise apply the extension step with $A = A_{n-1}$ and this value of n. Set

$$D = \bigcup_{n \geq 0} A_n.$$

Every positive integer n belongs to $\Delta(A_n)$, hence is a difference of two elements of D. Such a representation is unique: any two competing representations would have all their entries in some finite A_N, contradicting the distinct-difference property of A_N. Reversing the ordered pair proves existence and uniqueness for negative integers as well.

At every extension step $M \geq 27$, so

$$u = 2M + n + 1 \geq 56 > 38.$$

Thus no new element at most 38 is ever introduced, proving the first assertion. Finally, D is infinite, since a finite set has only finitely many differences. □

Lemma 3. *For $k \in \mathbb{Z}$, define*

$$L_k = D + k = \{d + k : d \in D\}.$$

Regard the integers as points and the sets L_k as lines. Then:

1. *every two distinct points lie on exactly one line;*
2. *every two distinct lines meet in exactly one point;*
3. *every line is infinite;*
4. *integer translations preserve the collection of lines.*

Proof. Let $p \neq q$ be integers. By Lemma 2, there is a unique ordered pair $d, e \in D$ satisfying

$$p - q = d - e.$$

Then

$$k = p - d = q - e$$

is the unique integer for which $p, q \in L_k$. This proves the first assertion.

For $k \neq \ell$, a point of $L_k \cap L_\ell$ corresponds precisely to an ordered pair $d, e \in D$ satisfying

$$k + d = \ell + e, \qquad \text{equivalently} \qquad d - e = \ell - k.$$

There is exactly one such ordered pair, so the intersection consists of exactly one point. Since each L_k is infinite, this also shows that different indices give different lines.

The third assertion follows from the infinitude of D, and the fourth from

$$L_k + s = L_{k+s}.$$

□

For distinct integers m, n, write $L(m, n)$ for their unique common line.

5 Construction and verification of the hyperfield

Let

$$\mathbb{H} = \{0\} \sqcup \{t^n : n \in \mathbb{Z}\},$$

with multiplication

$$t^m t^n = t^{m+n}, \qquad 0a = a0 = 0.$$

For distinct nonzero elements $a = t^m$ and $b = t^n$, define their line by

$$\Lambda(a, b) = \{t^r : r \in L(m, n)\} \subseteq \mathbb{H}^\times.$$

The incidence properties of Lemma 3 hold equally for these lines in $\mathbb{H}^\times$.

Define hyperaddition by

$$0 \boxplus a = a \boxplus 0 = \{a\}$$

and, for $a, b \neq 0$, by

$$a \boxplus b = \begin{cases} \{0, a\}, & a = b, \\ \Lambda(a, b) \setminus \{a, b\}, & a \neq b. \end{cases} \tag{1}$$

Proposition 4. *These operations make $\mathbb{H}$ a commutative hyperfield. Its additive negation is the identity map.*

Proof. The nonzero elements form an infinite cyclic multiplicative group with identity $1 = t^0$. Zero is absorbing by definition.

Nonempty sums, commutativity, and the neutral element. Every line is infinite, so all sums in (1) are nonempty. Commutativity and the additive neutral-element axiom follow directly from the definition.

Additive inverses and reversibility. For all $a, b \in \mathbb{H}$, the definition gives

$$0 \in a \boxplus b \quad \Longleftrightarrow \quad a = b. \tag{2}$$

Consequently, every element has a unique additive inverse, namely

$$-a = a.$$

To verify reversibility, first suppose that a, b, c are all nonzero. Then $c \in a \boxplus b$ holds precisely when either all three elements coincide, or they are pairwise distinct and lie on a common line. Both conditions are symmetric in a, b, c. If at least one element is zero, membership holds precisely when the other two elements are equal, again a symmetric condition. Hence

$$c \in a \boxplus b \quad \Longleftrightarrow \quad b \in c \boxplus a \quad \Longleftrightarrow \quad b \in c \boxplus (-a),$$

as required.

Associativity. If one of the three elements is zero, associativity follows from the neutral-element rule. We therefore consider nonzero elements.

If all three elements equal a, then

$$(a \boxplus a) \boxplus a = \{0, a\} \boxplus a = \{0, a\}.$$

Suppose next that the multiset of the three elements is $\{a, a, b\}$, where $a \neq b$, and let $\Lambda = \Lambda(a, b)$. Then

$$(a \boxplus a) \boxplus b = \{b\} \cup \big(\Lambda \setminus \{a, b\}\big) = \Lambda \setminus \{a\}.$$

Also, uniqueness of the line through two distinct points gives

$$\begin{aligned}(a \boxplus b) \boxplus a &= \bigcup_{s \in \Lambda \setminus \{a,b\}} \big(\Lambda \setminus \{s, a\}\big) \\ &= \Lambda \setminus \{a\}.\end{aligned}$$

For the last equality, given $v \in \Lambda \setminus \{a\}$, choose $s \in \Lambda \setminus \{a, b, v\}$, which is possible because Λ is infinite.

Now suppose that a, b, c are pairwise distinct and lie on one line Λ. We claim that

$$(a \boxplus b) \boxplus c = \{0\} \cup \Lambda. \tag{3}$$

Every summand on the left is contained in $\{0\} \cup \Lambda$. Choosing the intermediate element $c \in a \boxplus b$ produces both 0 and c. Given $v \in \Lambda \setminus \{c\}$, choose

$$s \in \Lambda \setminus \{a, b, c, v\}.$$

Then $s \in a \boxplus b$ and $v \in s \boxplus c$, proving (3).

Finally, suppose that a, b, c are noncollinear. In this case

$$(a \boxplus b) \boxplus c = \mathbb{H}^\times \setminus \big(\Lambda(a,b) \cup \Lambda(a,c) \cup \Lambda(b,c)\big). \tag{4}$$

To prove the forward inclusion, take

$$s \in \Lambda(a,b) \setminus \{a,b\}.$$

Since $c \notin \Lambda(a,b)$, the elements s, c are distinct. The line $\Lambda(s,c)$ meets $\Lambda(a,b)$ only at s, and meets each of $\Lambda(a,c)$ and $\Lambda(b,c)$ only at c. Therefore

$$s \boxplus c = \Lambda(s,c) \setminus \{s,c\}$$

avoids all three lines on the right of (4), as well as the hyperfield zero.

Conversely, let v lie outside those three lines. The line $\Lambda(c,v)$ meets $\Lambda(a,b)$ in a unique point s. This point is neither a nor b, since otherwise v would belong to $\Lambda(a,c)$ or $\Lambda(b,c)$. Also $v \neq s, c$. Thus

$$s \in a \boxplus b, \qquad v \in s \boxplus c,$$

proving the reverse inclusion.

The formulas just established are independent of the order in which the three elements are listed. Together with commutativity, they imply

$$(a \boxplus b) \boxplus c = a \boxplus (b \boxplus c)$$

in every case. Associativity for subsets follows by taking unions over their elements; hence finite hypersums can be parenthesized freely.

Distributivity. Multiplication by t^k acts on nonzero elements by translating their exponents by k. By Lemma 3, it carries lines to lines, and in particular

$$t^k \Lambda(t^m, t^n) = \Lambda(t^{m+k}, t^{n+k}) \qquad (m \neq n).$$

It follows from (1), including its equal-argument and zero-argument cases, that

$$t^k(a \boxplus b) = t^k a \boxplus t^k b.$$

For multiplication by zero, both sides are $\{0\}$, since $a \boxplus b$ is nonempty. Thus distributivity holds for all scalars. All hyperfield axioms have now been verified. □

6 Failure of the cubic implication

Proof of Theorem 1. Use the hyperfield constructed above and set

$$x = 1, \qquad y = t, \qquad z = t^6.$$

The integers $0, 1, 6$ all belong to D. Consequently,

$$t^6 \in \Lambda(1,t) \setminus \{1,t\} = 1 \boxplus t.$$

It follows that

$$0 \in t^6 \boxplus t^6 \subseteq (1 \boxplus t) \boxplus t^6.$$

Thus the hypothesis in the question is satisfied.

Consider the line

$$\Lambda = \{t^r : r \in D - 20\}.$$

Because $20, 23, 27 \in D$, this line contains

$$1, \qquad t^3, \qquad t^7.$$

On the other hand,

$$t^{18} \notin \Lambda,$$

because $18 \in D - 20$ would imply $38 \in D$, contrary to Lemma 2.

Write

$$u = 1, \qquad v = t^3, \qquad w = t^{18}, \qquad q = t^7, \qquad A = u \boxplus v \boxplus w.$$

The points u, v, w are noncollinear, and $\Lambda(u, v) = \Lambda$. Formula (4) therefore gives

$$A \subseteq \mathbb{H}^\times \setminus \Lambda.$$

In particular,

$$0 \notin A, \qquad q \notin A. \tag{5}$$

The repeated sum of q satisfies

$$q \boxplus q \boxplus q = \{0, q\} \boxplus q = \{0, q\}.$$

Hence

$$\begin{aligned} u \boxplus v \boxplus w \boxplus q \boxplus q \boxplus q &= A \boxplus \{0, q\} \\ &= A \cup \bigcup_{a \in A} (a \boxplus q). \end{aligned}$$

The first set on the last line does not contain zero by (5). By (2), a set $a \boxplus q$ contains zero exactly when $a = q$; this is also excluded by (5). Therefore

$$0 \notin 1 \boxplus t^3 \boxplus t^{18} \boxplus t^7 \boxplus t^7 \boxplus t^7.$$

Finally, multiplication and negation in this hyperfield give

$$x^3 = 1, \qquad y^3 = t^3, \qquad z^3 = t^{18}, \qquad -xyz = xyz = t^7.$$

Thus the last displayed nonmembership is precisely the failure of the conclusion asked about in the question. □

References

[1] A. Connes and C. Consani, *The hyperring of adèle classes*, Journal of Number Theory **131** (2011), no. 2, 159–194.